

Friction and Circular Motion

The forces that resist sliding and the forces that bend a path into a curve, from car tyres and blood cells to centrifuges.

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Chapter 3 mostly ignored friction and dealt with straight-line motion. Real engineering problems need both. Friction lets us walk, drive and grip, and it wastes energy in every machine. Air and fluid drag set the top speed of a skydiver and the settling rate of blood cells. Circular motion is everywhere: wheels, turbines, satellites, centrifuges, cars on curves. This chapter adds these two ingredients to Newton's laws. Along the way you will see that “centripetal force” is not a new kind of force, but a *role* that real forces play.

YOU WILL BE ABLE TO

- Distinguish static from kinetic friction, and use $f_s \leq \mu_s N$ and $f_k = \mu_k N$ correctly.
- Solve friction problems on level surfaces and inclines, including stacked blocks.
- Model drag forces (linear and quadratic) and calculate terminal velocities.
- Derive the centripetal acceleration $a_c = v^2/r = \omega^2 r$.
- Identify which real forces provide the centripetal force in flat and banked curves, conical pendulums and vertical circles.
- Handle non-uniform circular motion (tangential plus radial acceleration).
- Use centrifugal pseudo-forces correctly in rotating frames.

THINK FIRST

To stop as quickly as possible on a dry road, should a driver lock the wheels so that the tyres skid, or brake just hard enough that the wheels keep turning?

Keep the wheels turning. A rolling tyre grips the road through **static** friction (the contact patch is momentarily at rest on the road), and the static coefficient is larger than the kinetic coefficient for sliding. A skidding tyre is limited by the smaller **kinetic** friction, and it also cannot steer. Anti-lock braking systems (ABS) pulse the brakes many times a second to keep each wheel just short of locking. This gives near-maximum static friction *and* keeps steering control.

§4.1

Static and kinetic friction

When two surfaces are pressed together, a force parallel to the surfaces resists their sliding over each other. Microscopically, even polished surfaces are rough. They touch only at the tips of tiny peaks (asperities), where atoms bond temporarily. The true contact area is a tiny fraction of the apparent area, and it grows in proportion to how hard the surfaces are pressed together. That is why friction is proportional to the normal force, but nearly independent of the apparent contact area.

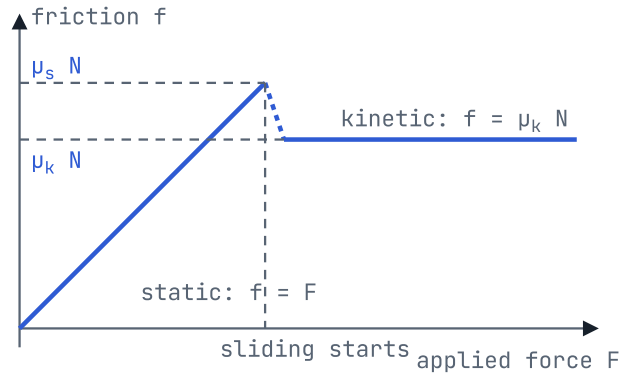
Static friction acts when there is no relative sliding. It adjusts its size and direction to prevent sliding, up to a maximum:

$$f_s \leq \mu_s N.$$

Kinetic friction acts when the surfaces slide. It points opposite to the *relative* velocity, and it is approximately constant:

$$f_k = \mu_k N.$$

Here N is the normal force, and μ_s and μ_k are the dimensionless **coefficients of friction**. Usually $\mu_k < \mu_s$.



Friction while the push on a resting block increases. Static friction matches the push until it reaches its limit $\mu_s N$. The block then slides, and friction drops to $\mu_k N$.

SURFACES	μ_s	μ_k
Rubber on dry concrete	1.0	0.8
Rubber on wet concrete	0.7	0.5
Steel on steel (dry)	0.74	0.57
Wood on wood	0.5	0.3
Ice on ice	0.1	0.03
Teflon on steel	0.04	0.04
Synovial joint (cartilage on cartilage)	0.01	0.003

COMMON MISTAKE

$f_s = \mu_s N$ is **not** a formula for static friction. It gives only the *maximum*. Static friction takes whatever value, from zero up to $\mu_s N$, keeps the surfaces from slipping. Find it from Newton's laws, then check that it does not exceed $\mu_s N$.

WORKED EXAMPLE 4.1

Will the box move?

A 20 kg box rests on a floor with $\mu_s = 0.50$ and $\mu_k = 0.40$. A worker pushes it horizontally with (a) 80 N and (b) 120 N. Find the friction force and the acceleration in each case.

Solution

The normal force is $N = mg = 196$ N. So the maximum static friction is $\mu_s N = 98$ N, and the kinetic friction is $\mu_k N = 78.4$ N.

(a) $80 \text{ N} < 98 \text{ N}$, so static friction can hold the box. It adjusts to exactly $f_s = 80$ N, opposite to the push, and $a = 0$.

(b) $120 \text{ N} > 98 \text{ N}$, so the box slides. Kinetic friction acts:

$$a = \frac{F - \mu_k N}{m} = \frac{120 - 78.4}{20} = 2.08 \text{ m/s}^2.$$

Evaluate. Once the box is moving, the worker could ease off to just over 78.4 N and keep it sliding at constant speed. Starting is harder than continuing because $\mu_s > \mu_k$.

Friction on an incline

A block on an incline at angle θ has $N = mg \cos \theta$, provided nothing else pushes perpendicular to the surface. The component of gravity down the slope is $mg \sin \theta$. The block stays at rest if $mg \sin \theta \leq \mu_s mg \cos \theta$, that is, if

$$\tan \theta \leq \mu_s.$$

The steepest angle at which it can rest is the **angle of repose**, $\theta_r = \tan^{-1} \mu_s$. Tilting a surface until an object just slips is a standard way to measure μ_s .

WORKED EXAMPLE 4.2

Up and down a rough ramp

A 5.0 kg block is on a 30° incline with $\mu_k = 0.20$ and $\mu_s = 0.30$. (a) Find its acceleration when it slides down. (b) It is instead given a push so that it starts *up* the slope at 6.0 m/s. How far up does it go? (c) Does it then slide back down?

Solution

Use axes along the slope, with $N = mg \cos 30^\circ = 42.4$ N.

(a) **Sliding down.** Friction acts up the slope:

$$a = g(\sin 30^\circ - \mu_k \cos 30^\circ) = 9.8(0.500 - 0.173) = 3.20 \text{ m/s}^2 \text{ down the slope.}$$

(b) **Sliding up.** Now friction also acts **down** the slope, opposing the motion. Gravity and friction both slow the block:

$$a = -g(\sin 30^\circ + \mu_k \cos 30^\circ) = -9.8(0.673) = -6.60 \text{ m/s}^2.$$

From $v^2 = u^2 + 2as$ with $v = 0$:

$$s = \frac{u^2}{2|a|} = \frac{36}{13.2} = 2.73 \text{ m.}$$

(c) At rest at the top, the block slides back only if $\tan 30^\circ = 0.577$ exceeds $\mu_s = 0.30$. It does, so the block **slides back down**, now with $a = 3.20 \text{ m/s}^2$ from part (a).

Evaluate. The deceleration going up (6.60) is larger than the acceleration coming down (3.20), because friction always opposes the motion. So the trip up is quicker than the trip down, and the block returns to its starting point slower than 6.0 m/s.

WORKED EXAMPLE 4.3

The best angle to pull

You drag a 20 kg crate across a floor ($\mu_k = 0.50$) with a rope at angle θ above the horizontal, at constant velocity. Find the tension needed as a function of θ , and the angle that minimises it.

Solution

FBD. Four forces act: the tension T at θ , the weight mg , the normal force N , and kinetic friction $\mu_k N$ backward. Lifting slightly reduces N , and therefore the friction.

$$\text{Vertical: } N = mg - T \sin \theta, \quad \text{Horizontal: } T \cos \theta = \mu_k N = \mu_k (mg - T \sin \theta).$$

Solving for T :

$$T(\theta) = \frac{\mu_k mg}{\cos \theta + \mu_k \sin \theta}.$$

Minimise. T is smallest when the denominator is largest. Set its derivative to zero:

$$-\sin \theta + \mu_k \cos \theta = 0, \text{ so}$$

$$\tan \theta^* = \mu_k \Rightarrow \theta^* = \tan^{-1} 0.50 = 26.6^\circ.$$

At this angle the denominator equals $\sqrt{1 + \mu_k^2}$, so

$$T_{\min} = \frac{\mu_k mg}{\sqrt{1 + \mu_k^2}} = \frac{0.50(196)}{1.118} = 87.7 \text{ N.}$$

Pulling horizontally would need $\mu_k mg = 98 \text{ N}$.

Evaluate. Pulling slightly upward saves about 10% of the effort. Pulling too steeply wastes effort lifting rather than dragging.

IN PRACTICE

Friction in engineering and in the body. Engineers sometimes need *more* friction (tyres, brake pads, belt drives, the soles of shoes) and sometimes much *less* (bearings, piston rings, artificial joints). Tyre treads channel water away so the rubber can grip the road. At high speed on a wet road, a film of water can lift the tyre completely, a dangerous loss of friction called aquaplaning. Human synovial joints are astonishingly slippery: cartilage lubricated by synovial fluid reaches $\mu \approx 0.003$, better than most engineered bearings. Osteoarthritis degrades this surface, and part of the engineering challenge of hip and knee replacements is to find material pairs (polished cobalt–chromium or ceramic against ultra-high-molecular-weight polyethylene) that keep friction and wear low for decades.

§4.2

Drag forces and terminal velocity

A body moving through a fluid (air, water, blood) feels a **drag force** opposite to its velocity relative to the fluid. Unlike sliding friction, drag depends strongly on speed. Two limiting models cover most cases.

- **Linear (viscous) drag**, for small, slow objects in viscous fluids (cells, dust, droplets):

$$D = bv.$$

For a sphere of radius r in a fluid of viscosity η , Stokes' law gives $b = 6\pi\eta r$.

- **Quadratic drag**, for larger, faster objects (people, cars, balls):

$$D = \frac{1}{2} C_p A v^2.$$

Here ρ is the fluid density, A the cross-sectional area facing the flow, and C the drag coefficient: about 0.5 for a sphere, about 1.0 for a skydiver spread flat, and about 0.3 for a modern car.

The Reynolds number of Chapter 1 decides which model applies: linear drag when $Re \ll 1$, quadratic drag when $Re \gg 1000$.

A falling body accelerates until drag balances gravity (and buoyancy, if it matters). It then falls at constant terminal velocity v_t :

$$\text{quadratic: } mg = \frac{1}{2}C\rho A v_t^2 \Rightarrow v_t = \sqrt{\frac{2mg}{C\rho A}}; \quad \text{linear: } mg_{\text{eff}} = b v_t \Rightarrow v_t = \frac{mg_{\text{eff}}}{b}.$$

WORKED EXAMPLE 4.4

A skydiver's terminal speed

An 80 kg skydiver falls spread-eagled, with $A = 0.70 \text{ m}^2$ and $C = 1.0$, through air of density 1.2 kg/m^3 . Find the terminal speed. What does it become if she dives head-first with $A = 0.18 \text{ m}^2$ and $C = 0.7$?

Solution

Spread-eagled:

$$v_t = \sqrt{\frac{2(80)(9.8)}{1.0(1.2)(0.70)}} = \sqrt{\frac{1568}{0.84}} = \sqrt{1867} = 43 \text{ m/s (155 km/h)}.$$

Head-first:

$$v_t = \sqrt{\frac{1568}{0.7(1.2)(0.18)}} = \sqrt{\frac{1568}{0.151}} = \sqrt{10370} = 102 \text{ m/s } (\approx 370 \text{ km/h}).$$

Evaluate. Because $v_t \propto 1/\sqrt{CA}$, cutting CA by a factor of 5.6 raises the terminal speed by $\sqrt{5.6} = 2.4$. Speed skydivers exploit exactly this. A parachute increases CA about 25-fold, cutting the terminal speed by a factor of 5, to about 5–9 m/s.

WORKED EXAMPLE 4.5

Sedimentation of red blood cells

The erythrocyte sedimentation rate (ESR) is a routine blood test: whole blood stands in a vertical tube, and the drop of the boundary between red cells and plasma is measured in mm per hour. Model a red cell as a sphere of radius $4.0 \text{ }\mu\text{m}$ and density 1100 kg/m^3 , settling in

plasma of density 1025 kg/m^3 and viscosity $1.2 \times 10^{-3} \text{ Pa s}$. Find its terminal speed in mm/h. How would a centrifuge producing $1000 g$ change this?

Solution

Forces on a settling cell: weight $\rho_c V g$ down, buoyancy $\rho_p V g$ up (Archimedes, Chapter 10) and Stokes drag $6\pi\eta r v$ up, with $V = \frac{4}{3}\pi r^3$. At terminal velocity they balance:

$$(\rho_c - \rho_p) \frac{4}{3}\pi r^3 g = 6\pi\eta r v_t \Rightarrow v_t = \frac{2r^2(\rho_c - \rho_p)g}{9\eta}.$$

Substitute:

$$v_t = \frac{2(4.0 \times 10^{-6})^2(75)(9.8)}{9(1.2 \times 10^{-3})} = \frac{2.35 \times 10^{-8}}{1.08 \times 10^{-2}} = 2.2 \times 10^{-6} \text{ m/s}.$$

Converting: $2.2 \times 10^{-6} \text{ m/s} \times 3600 \text{ s/h} = 7.8 \times 10^{-3} \text{ m/h}$, about 8 mm/h.

With a centrifuge, g is replaced by $1000 g$, so $v_t \approx 2.2 \text{ mm/s}$ and separation takes seconds to minutes instead of hours.

Evaluate. Normal ESR values are roughly 0–20 mm/h, so the simple model is the right order of magnitude. In inflammation, plasma proteins make red cells stack into columns called *rouleaux*. A stack behaves like a larger particle, and since $v_t \propto r^2$, it settles much faster. That is why a raised ESR signals inflammation.

§4.3

Uniform circular motion

A particle moving round a circle at **constant speed** is accelerating, because the direction of its velocity keeps changing.

Angular description. If the particle moves through an angle θ (in radians) on a circle of radius r , the arc length is $s = r\theta$. The **angular velocity** is $\omega = d\theta/dt$ (rad/s), and the speed is

$$v = r\omega.$$

The **period** is $T = 2\pi r/v = 2\pi/\omega$, and the **frequency** is $f = 1/T$, so $\omega = 2\pi f$.

Deriving the centripetal acceleration. Put the circle at the origin, with $\theta = \omega t$:

$$\vec{r} = r \cos \omega t \hat{i} + r \sin \omega t \hat{j}.$$

Differentiate twice:

$$\vec{v} = \frac{d\vec{r}}{dt} = -r\omega \sin \omega t \hat{i} + r\omega \cos \omega t \hat{j}, \quad \vec{a} = \frac{d\vec{v}}{dt} = -\omega^2 (r \cos \omega t \hat{i} + r \sin \omega t \hat{j}) = -\omega^2 \vec{r}.$$

The velocity is perpendicular to $\vec{r}$ (tangent to the circle), with magnitude $r\omega$. The acceleration points towards the centre, opposite to $\vec{r}$, with magnitude

$$a_c = \omega^2 r = \frac{v^2}{r}.$$

KEY IDEA

Centripetal acceleration. A body moving in a circle of radius r at speed v has an acceleration $v^2/r = \omega^2 r$ directed towards the centre. By Newton's second law, the net force towards the centre must be

$$\sum F_{\text{radial}} = \frac{mv^2}{r}.$$

“Centripetal force” is not a new kind of force. It is whatever combination of real forces (tension, gravity, normal force, friction) points towards the centre. Never add an extra “centripetal force” to a free-body diagram.

§4.4

Dynamics of circular motion

The method is always the same. Draw the FBD, take one axis **towards the centre** of the circle, and set the net force along it equal to mv^2/r .

A car on a flat curve

On a level road, only **static friction** between the tyres and the road can push the car towards the centre. (The contact patch does not slide sideways, so the friction is static.) The requirement is $f_s = mv^2/r \leq \mu_s mg$, so the maximum safe speed is

$$v_{\text{max}} = \sqrt{\mu_s g r}.$$

This is independent of the car's mass.

WORKED EXAMPLE 4.6

Cornering on a dry and a wet road

Find the maximum speed for a car rounding a flat curve of radius 50 m when the road is (a) dry ($\mu_s = 0.80$) and (b) wet ($\mu_s = 0.40$).

Solution

(a) Dry:

$$v_{\max} = \sqrt{0.80(9.8)(50)} = \sqrt{392} = 19.8 \text{ m/s (71 km/h)}.$$

(b) Wet:

$$v_{\max} = \sqrt{0.40(9.8)(50)} = \sqrt{196} = 14.0 \text{ m/s (50 km/h)}.$$

Evaluate. Halving the friction coefficient cuts the safe speed by a factor of $\sqrt{2}$, not 2. A driver who takes this bend at 70 km/h in the rain will skid outward along the tangent: not “thrown outward”, but simply continuing in a straight line once friction can no longer bend the path.

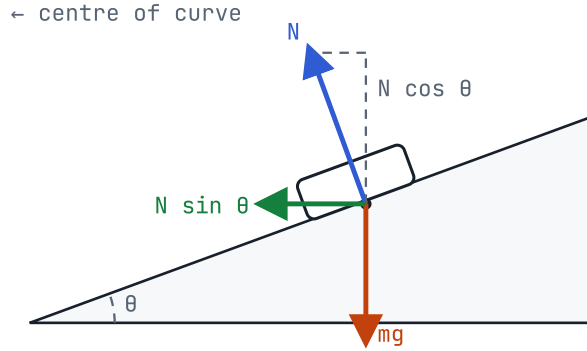
Banked curves

Banking the road tilts the normal force towards the centre, so that part of it provides the centripetal force. With no friction at all, the forces on the car are N (perpendicular to the road surface) and mg :

$$\text{Vertical: } N \cos \theta = mg, \quad \text{Radial: } N \sin \theta = \frac{mv^2}{r}.$$

Dividing the equations gives the **design speed** at which no friction is needed:

$$\tan \theta = \frac{v^2}{gr}.$$



On a banked road, the horizontal component of the normal force, $N \sin \theta$, supplies the centripetal force. $N \cos \theta$ balances the weight.

If the car goes faster than the design speed, friction acts **down** the slope to help hold it in. If it goes slower, friction acts **up** the slope to stop it sliding down. Including friction, the range of safe speeds is

$$v_{\max}^2 = gr \frac{\tan \theta + \mu_s}{1 - \mu_s \tan \theta}, \quad v_{\min}^2 = gr \frac{\tan \theta - \mu_s}{1 + \mu_s \tan \theta}.$$

If $\mu_s \geq \tan \theta$, there is no minimum speed: a parked car will not slide down.

WORKED EXAMPLE 4.7

Designing a banked curve

A highway curve of radius 100 m is to be banked so that cars at 20 m/s (72 km/h) need no friction. (a) Find the banking angle. (b) If $\mu_s = 0.30$, find the range of speeds at which a car can take the curve without slipping.

Solution

(a) Design angle:

$$\tan \theta = \frac{v^2}{gr} = \frac{400}{980} = 0.408 \Rightarrow \theta = 22.2^\circ.$$

(b) Maximum speed, with friction acting down the slope:

$$v_{\max}^2 = 980 \cdot \frac{0.408 + 0.30}{1 - (0.30)(0.408)} = 980 \cdot \frac{0.708}{0.878} = 790 \Rightarrow v_{\max} = 28.1 \text{ m/s (101 km/h)}.$$

Minimum speed, with friction acting up the slope:

$$v_{\min}^2 = 980 \cdot \frac{0.408 - 0.30}{1 + (0.30)(0.408)} = 980 \cdot \frac{0.108}{1.122} = 94.3 \Rightarrow v_{\min} = 9.7 \text{ m/s (35 km/h)}.$$

Evaluate. Banking widens the safe window a great deal. On a flat road with the same friction, the limit would be $\sqrt{0.30(9.8)(100)} = 17 \text{ m/s}$. Real highways use smaller angles (typically 4° – 8°), because vehicles must be able to stop on an icy curve without sliding sideways. Racing tracks and velodromes bank much more steeply.

Vertical circles

In a vertical circle, gravity has a component along the radius that changes around the circle, so the speed changes. Even so, at the top and the bottom the radial equation is simple. For a ball on a string of length L :

$$\text{Top: } T_{\text{top}} + mg = \frac{mv_{\text{top}}^2}{L}, \quad \text{Bottom: } T_{\text{bottom}} - mg = \frac{mv_{\text{bottom}}^2}{L}.$$

The string stays taut at the top only if $T_{\text{top}} \geq 0$, which requires

$$v_{\text{top}} \geq \sqrt{gL}.$$

The same condition, $v \geq \sqrt{gR}$, keeps a roller-coaster car on the track at the top of a loop. A car going over a hump of radius R leaves the road if $v > \sqrt{gR}$.

WORKED EXAMPLE 4.8

Whirling a ball in a vertical circle

A 0.20 kg ball on a 0.80 m string is whirled in a vertical circle. (a) Find the minimum speed at the top for the string to stay taut. (b) If the ball has this speed at the top, its speed at the bottom is $v_b = \sqrt{v_t^2 + 4gL}$ (we prove this with energy conservation in Chapter 5). Find v_b and the tension at the bottom.

Solution

(a) Minimum speed at the top:

$$v_t = \sqrt{gL} = \sqrt{9.8(0.80)} = 2.80 \text{ m/s}.$$

(b) At the bottom:

$$v_b^2 = gL + 4gL = 5gL = 39.2 \text{ m}^2/\text{s}^2 \Rightarrow v_b = 6.26 \text{ m/s}.$$

$$T_b = mg + \frac{mv_b^2}{L} = mg + 5mg = 6mg = 6(0.20)(9.8) = 11.8 \text{ N.}$$

Evaluate. The tension at the bottom is six times the ball's weight. That makes the bottom of a vertical circle the place where strings snap, and where pilots pulling out of a dive feel the largest load.

IN PRACTICE

g-forces on pilots and patients. A fighter pilot pulling out of a dive at 250 m/s on a circle of radius 1000 m experiences $v^2/r = 62.5 \text{ m/s}^2$ of centripetal acceleration, which, added to gravity, gives an apparent weight about **7.4** times normal (“7.4 g”). Blood is driven towards the feet, and the heart may be unable to pump enough of it up to the brain. Vision greys out, and then the pilot loses consciousness (G-LOC). Anti-g suits squeeze the legs and abdomen to resist this pooling. The same physics, run in reverse, is used in human centrifuges to train pilots and astronauts and to study cardiovascular responses.

The conical pendulum

A bob on a string of length L moving in a horizontal circle, with the string at angle θ to the vertical, is a **conical pendulum**. The circle has radius $r = L \sin \theta$.

$$\text{Vertical: } T \cos \theta = mg, \quad \text{Radial: } T \sin \theta = m\omega^2 L \sin \theta.$$

The second equation gives $T = m\omega^2 L$. Dividing the two equations gives

$$\omega^2 = \frac{g}{L \cos \theta}, \quad \text{period } P = 2\pi \sqrt{\frac{L \cos \theta}{g}}.$$

WORKED EXAMPLE 4.9

A conical pendulum

A 0.30 kg bob on a 1.0 m string moves in a horizontal circle with the string at 30° to the vertical. Find the period, the speed and the tension.

Solution

Period:

$$P = 2\pi \sqrt{\frac{1.0 \cos 30^\circ}{9.8}} = 2\pi \sqrt{0.0884} = 1.87 \text{ s.}$$

Speed: the radius is $r = 1.0 \sin 30^\circ = 0.50$ m, so

$$v = \frac{2\pi r}{P} = \frac{2\pi(0.50)}{1.87} = 1.68 \text{ m/s.}$$

Tension:

$$T = \frac{mg}{\cos 30^\circ} = \frac{0.30(9.8)}{0.866} = 3.39 \text{ N.}$$

Check: $T \sin 30^\circ = 1.70$ N, and $mv^2/r = 0.30(1.68)^2/0.50 = 1.69$ N. ✓

§4.5

Non-uniform circular motion

If the speed changes along a circular path, the acceleration has two perpendicular components:

- a **tangential** component, $a_t = dv/dt = r\alpha$, which changes the speed, where $\alpha = d\omega/dt$ is the angular acceleration;
- a **radial** (centripetal) component, $a_c = v^2/r$, which changes the direction.

The total acceleration has magnitude

$$a = \sqrt{a_t^2 + a_c^2}.$$

The same split applies to *any* curved path, with r replaced by the local radius of curvature. This is how engineers analyse a car that brakes while cornering. The tyres must provide the vector sum of both accelerations, and that sum must stay inside the “friction circle” of radius $\mu_s g$.

§4.6

Rotating frames and the centrifugal force

In a frame rotating with angular velocity ω , a body at rest at radius r appears to be in equilibrium. Newton’s second law can still be used in this frame if we add an outward pseudo-force, the **centrifugal force**:

$$F_{cf} = m\omega^2 r \quad (\text{directed away from the axis}).$$

A body that *moves* in the rotating frame also feels a sideways pseudo-force, the **Coriolis force**,

$-2m\vec{\omega} \times \vec{v}_{\text{rel}}$. On the rotating Earth, the Coriolis force sets the spin of hurricanes and deflects long-range projectiles.

COMMON MISTAKE

Use the centrifugal force **only** when you deliberately work in the rotating frame. In the ground frame there is no outward force on a car going round a bend. There is only an inward net force, and the passengers' sense of being "thrown outward" is their inertia carrying them straight on while the car turns beneath them.

IN PRACTICE

Centrifuges. A centrifuge's strength is quoted as its *relative centrifugal force*, $\text{RCF} = \omega^2 r / g$. At 10 000 rpm, $\omega = 10\,000 \times 2\pi/60 = 1047$ rad/s. At $r = 10$ cm, this gives $\omega^2 r = 1.1 \times 10^5$ m/s², about 11 000 g. In the tube's rotating frame, the sample feels an "effective gravity" that strong. By Example 4.5, red cells then settle about 10^4 times faster than on the bench. Ultracentrifuges reach 10^6 g and can separate proteins and viruses by mass. Uranium enrichment uses gas centrifuges on the same principle.

CHAPTER SUMMARY

- Static friction: $f_s \leq \mu_s N$, adjusting to prevent slipping. Kinetic friction: $f_k = \mu_k N$, opposite to the relative velocity. Usually $\mu_k < \mu_s$. Angle of repose: $\tan \theta_r = \mu_s$.
- Drag: $D = bv$ (Stokes: $b = 6\pi\eta r$) or $D = \frac{1}{2}C\rho A v^2$. Terminal velocity is reached when drag balances the effective weight.
- Uniform circular motion: $v = r\omega$, $T = 2\pi/\omega$, and $a_c = v^2/r = \omega^2 r$ towards the centre.
- The net inward force must equal mv^2/r . "Centripetal force" is a role played by real forces.
- Flat curve: $v_{\text{max}} = \sqrt{\mu_s g r}$. Banked curve with no friction: $\tan \theta = v^2/gr$.
- Vertical circle: $v_{\text{top}} \geq \sqrt{gR}$ to stay in contact. Tension is greatest at the bottom.
- Conical pendulum: $\omega^2 = g/(L \cos \theta)$.
- Non-uniform circular motion: $a = \sqrt{a_t^2 + (v^2/r)^2}$.
- Rotating frame: add the centrifugal force $m\omega^2 r$ outward (plus Coriolis for moving bodies).

Practice problems

Full step-by-step solutions are in the separate solutions PDF.

LEVEL A – CONCEPT CHECK

P4.1

Why is it usually easier to keep a heavy box sliding than to start it moving? Explain in terms of the microscopic picture of friction.

P4.2

“Friction always opposes motion.” Give two examples where friction acts in the *same* direction as the motion of the body it acts on, and explain what friction really opposes.

P4.3

A car rounds a curve at a constant 60 km/h. Is it accelerating? Which force provides the centripetal force? In the ground frame, is there an outward “centrifugal force” on the car?

P4.4

A roller-coaster car goes over the top of a vertical loop of radius R , upside down. At what speed does the rider feel weightless at the top? What does the rider feel if the car goes faster than this?

LEVEL B – STANDARD PROBLEMS

P4.5

A 2.0 kg block sits on a plank, and one end of the plank is slowly raised. The block starts to slide when the plank is at 35° . Once sliding at that angle, it accelerates at 2.0 m/s^2 . Find μ_s and μ_k .

P4.6

A crate rests on the flat bed of a truck, with $\mu_s = 0.40$ between crate and bed. (a) What is the greatest acceleration the truck can have without the crate sliding? (b) The truck is moving at 20 m/s. What is its shortest stopping distance if the crate is not to slide?

P4.7

A 3.0 kg block on a horizontal table ($\mu_k = 0.25$) is joined by a light string over an ideal pulley at the table's edge to a hanging 2.0 kg mass. Find the acceleration and the tension.

P4.8

A curve of radius 200 m is banked at 15° . At what speed can a car take the curve with no friction needed? Express your answer in km/h.

P4.9

A Ferris wheel of radius 15 m turns once every 30 s. Find the apparent weight of a 60 kg rider at the top and at the bottom.

P4.10

A 0.50 kg puck on a frictionless table is tied to a string through a hole at the centre, and moves in a circle of radius 1.2 m. The string breaks at a tension of 50 N. What is the maximum speed of the puck? Describe its path after the string breaks.

P4.11

A clinical centrifuge must produce 5000 g at a radius of 8.0 cm. At how many revolutions per minute must it spin?

P4.12

A 75 kg skydiver has a terminal speed of 60 m/s. (a) Find the drag constant k in $D = kv^2$. (b) The parachute increases k by a factor of 25. Find the new terminal speed. (c) If the parachute opened instantly at 60 m/s, what would the skydiver's deceleration be, in multiples of g ? Why are parachutes designed to open over a second or two?

LEVEL C – CHALLENGE PROBLEMS**P4.13**

A 1.0 kg block sits on top of a 4.0 kg block, which rests on a frictionless floor. Between the blocks, $\mu_s = 0.30$ and $\mu_k = 0.25$. A horizontal force F is applied to the lower block. (a) What is the largest F for which the blocks move together? (b) If $F = 25$ N, find the acceleration of each block.

P4.14

Derive the formula for the maximum safe speed on a banked curve with friction:

$$v_{\max}^2 = gr \frac{\tan \theta + \mu_s}{1 - \mu_s \tan \theta}.$$

Then find both $v_{\max}$ and $v_{\min}$ for $r = 60$ m, $\theta = 20^\circ$ and $\mu_s = 0.25$.

P4.15

The “Rotor” ride. Riders stand against the inside wall of a vertical cylinder of radius 2.5 m, which spins about its axis. When it reaches full speed, the floor drops away and the riders stay pinned to the wall. If $\mu_s = 0.40$ between clothing and wall, what is the minimum angular speed (in rad/s and in rpm)? Does the answer depend on the rider’s mass?

P4.16

A small particle of mass m is released from rest in a fluid and falls under gravity with linear drag $\mathbf{D} = b\mathbf{v}$ (ignore buoyancy). (a) Show that $\mathbf{v}(t) = v_t(1 - e^{-t/\tau})$, and identify v_t and τ . (b) How long does it take to reach 95% of terminal speed? (c) Evaluate τ for the red blood cell of Example 4.5, and comment on whether it is reasonable to assume the cell always moves at terminal speed.

P4.17

A small bead can slide without friction on a circular hoop of radius R that stands in a vertical plane and rotates about its vertical diameter with angular speed ω . Show that, besides the bottom of the hoop, the bead can be in equilibrium (relative to the hoop) at an angle θ from the bottom given by $\cos \theta = g/(\omega^2 R)$, but only if $\omega > \sqrt{g/R}$. Evaluate θ for $R = 0.20$ m at 120 rpm.

P4.18

A coin sits 10 cm from the centre of a turntable, with $\mu_s = 0.40$. The turntable starts from rest with a constant angular acceleration of 2.0 rad/s^2 . (a) Explain why the friction on the coin is not directed towards the centre. (b) Find the angular speed and the time at which the coin starts to slip.

All worked examples and practice problems in these notes are original.
Textbooks for further reading and practice, with a chapter-by-chapter map, are listed in the References.

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